

Lem Aut Disco Unidad Composicion

Involuciones

Lema 1. Sean $a, b \in \mathbb{D}$, entonces

$$\tau_b \circ \tau_a = \rho_\gamma \circ \tau_c \quad \text{donde} \quad c = \tau_a(b) = \frac{a-b}{1-\bar{a}b} \quad \wedge \quad \gamma = \begin{cases} \tau_{\bar{a}b}(1) & \text{si } a \neq b, \\ -1 & \text{si } a = b. \end{cases}$$

Demostración: Por el teo-aut-disco-unidad-parametros/teorema 1, sabemos que

$$c = (\tau_b \circ \tau_a)^{-1}(0) = (\tau_a^{-1} \circ \tau_b^{-1})(0) = \tau_a(\tau_b(0)) = \tau_a(b) = \frac{a-b}{1-\bar{a}b}.$$

Si $a = b$, entonces $c = 0$ y $\tau_b \circ \tau_a = \text{id} = \rho_{-1} \circ \tau_0$, luego $\gamma = -1$. Si $a \neq b$, entonces $(\tau_b \circ \tau_a)(0) = \rho_\gamma(c)$, luego $\tau_b(a) = \gamma \tau_a(b)$ y, como $a \neq b$, $\tau_a(b) \neq 0$, tenemos

$$\gamma = \frac{\tau_b(a)}{\tau_a(b)} = \frac{(b-a)/(1-\bar{b}a)}{(a-b)/(1-\bar{a}b)} = \frac{\bar{a}b-1}{1-\bar{a}b} = \tau_{\bar{a}b}(1).$$

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Referenciado en

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